

UNNS Color Confinement - Flux-Tube Extension Report

Route localization across increasing static-source separation

Milestone: Third formal report in the UNNS + color confinement investigation.

Status: Formal interpretation report based on Baker flux-tube extension, QC, and review-separately decisions.

Executive result

The Baker flux-tube extension gives a controlled separation-dependent dataset for the UNNS interpretation of color confinement as localized route geometry. After quality control, the main interpretable trend is restricted to ordinary FULL and NP transverse chromoelectric profiles over $d = 0.723\text{-}1.060$ fm, with 30 accepted rows.

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source separation d -> route-extension coordinate
E_x(x_t; d) -> localized route geometry
peak decrease + width increase -> route redistribution under extension
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This report does not claim a derived QCD confinement law. It formalizes a data-grounded structural mapping: the color route remains localized, while its profile changes under increasing static-source separation.

1. Inputs and data boundary

The full Baker extension parsed 1090 pointwise rows, representing 41 profiles from 10 source files. The QC pass then separated those rows into accepted, review-only, and excluded groups. The formal interpretation uses only ACCEPT_FOR_TREND rows from ordinary FULL / NP components over the trusted separation range.

2. Trusted trend dataset

component	n	d min	d max	peak min	peak max	FWHM min	FWHM max	area min	area max
FULL	15	0.723	1.060	0.195	0.387	0.530	0.663	0.159	0.231
NP	15	0.723	1.060	0.126	0.274	0.544	0.723	0.107	0.171

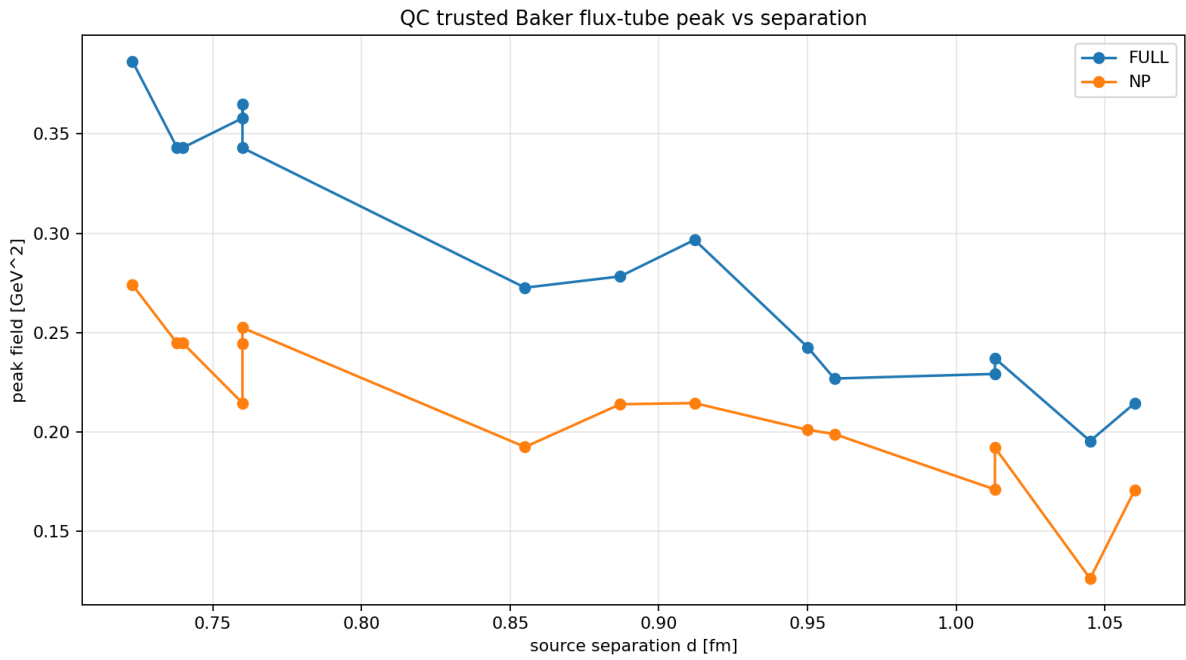
The trusted rows are sufficient for trend inspection, not for a final fitted physical law.

3. Trend diagnostics

The following linear summaries are descriptive diagnostics over median-by-distance trusted rows. They are not proposed as final law coefficients.

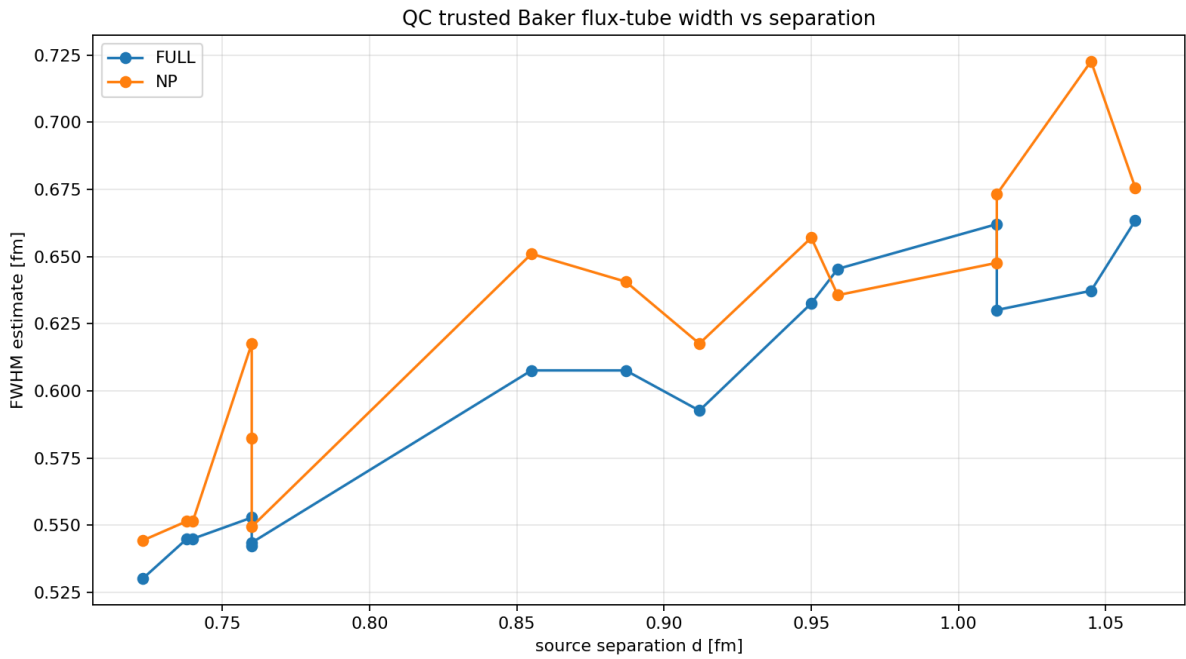
component	quantity	slope vs d	corr. r	n distances
FULL	peak	-0.4885	-0.960	12
FULL	fwhm	0.3711	0.964	12
FULL	area	-0.1919	-0.969	12
NP	peak	-0.2939	-0.908	12
NP	fwhm	0.4207	0.928	12
NP	area	-0.0818	-0.825	12

3.1 Trusted peak field trend



The trusted trend shows decreasing peak field with increasing separation for both FULL and NP profiles.

3.2 Trusted width trend



The trusted trend shows a modest increase in transverse width across the same interval.

4. Review-separately evidence class

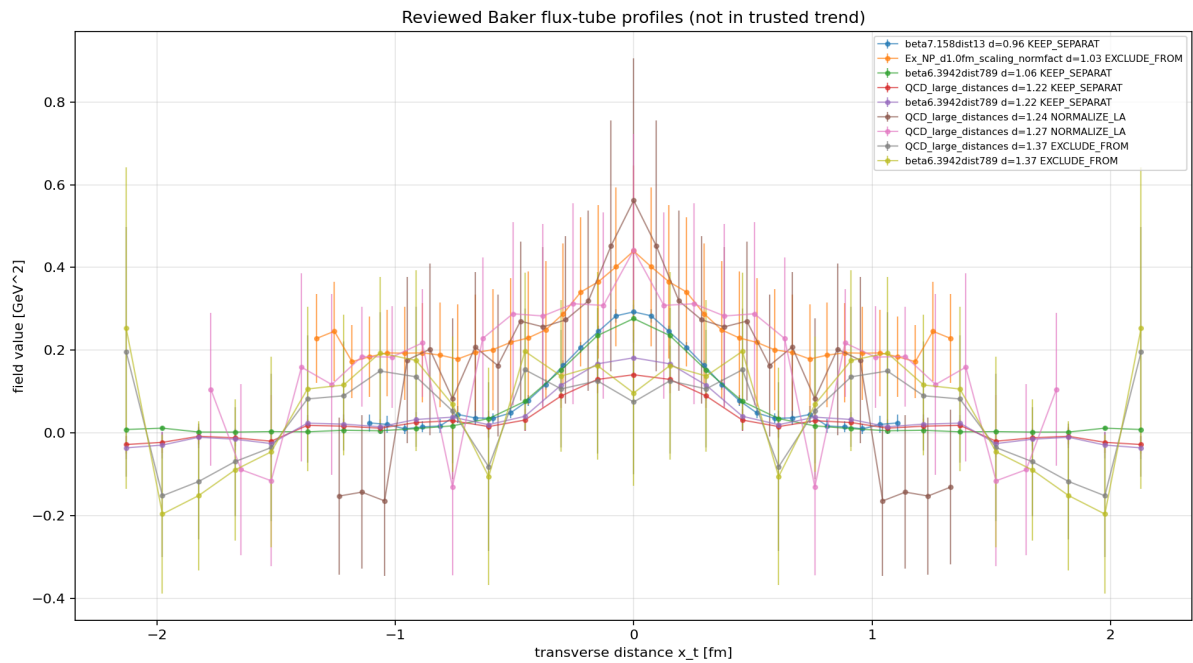
The fifth diagnostic inspected rows that QC marked REVIEW_SEPARATELY. These rows are valuable, but they do not belong in the main trend plot. Several are PAPER_DEFINED rather than ordinary FULL/NP scaling-normalized rows; others are large-distance profiles with lower signal-to-noise or unstable width estimates.

Main trend: ACCEPT_FOR_TREND rows only.
Separate qualitative string-breaking: beta7.158dist13.agr and beta6.3942dist789.agr at $d = 1.064$ and 1.216 fm.
Large-distance control: QCD_large_distances.agr at $d = 1.216$ fm.
Normalize later / qualitative only: QCD_large_distances.agr at $d = 1.235$ and 1.267 fm.
Excluded: Ex_NP $d = 1.033$ fm and the $d = 1.368$ fm rows.

Review decision table

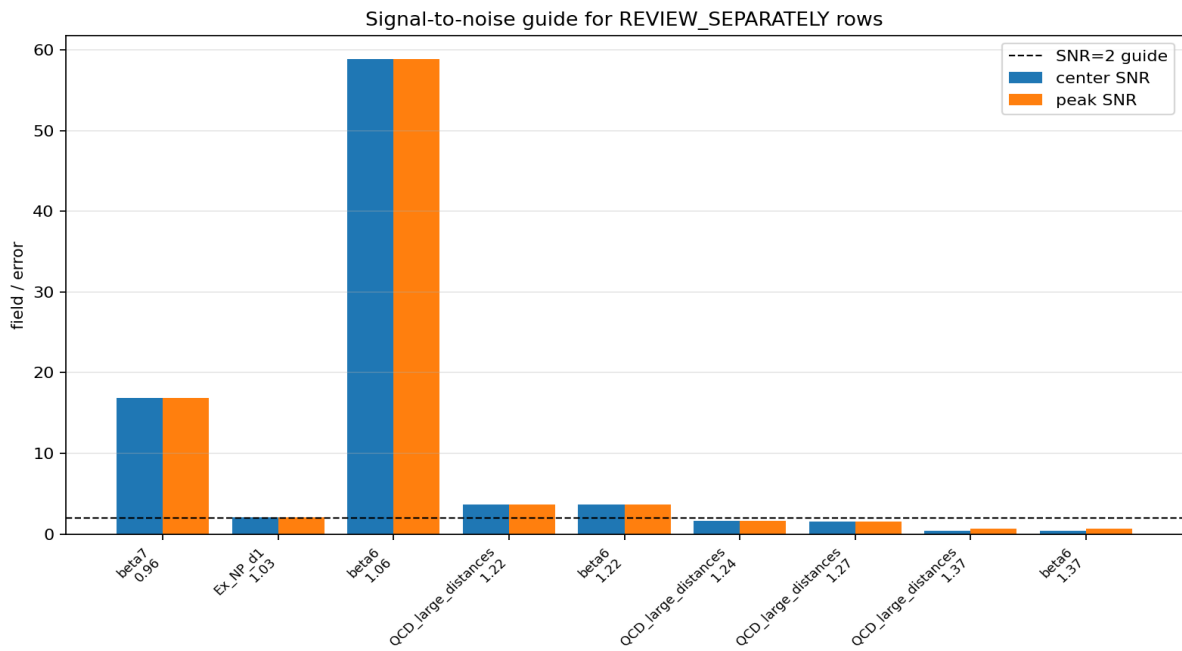
source file	comp	d	peak	SNR	FWHM	decision
beta7.158dist13.agr	PAPER_DEFINED	1.060	0.293	16.86	0.645	KEEP SEPARATE QUALITATIVE STRING BREAKING
Ex_NP_d1.0fm_scaling_normfact.agr	NP	1.033	0.440	2.12	2.458	EXCLUDE FROM TREND KEEP AS OUTLIER NOTE
beta6.3942dist789.agr	PAPER_DEFINED	1.064	0.277	58.84	0.663	KEEP SEPARATE QUALITATIVE STRING BREAKING
QCD_large_distances.agr	PAPER_DEFINED	1.216	0.141	3.67	0.708	KEEP SEPARATE LARGE DISTANCE CONTROL
beta6.3942dist789.agr	PAPER_DEFINED	1.216	0.182	3.67	0.708	KEEP SEPARATE QUALITATIVE STRING BREAKING
QCD_large_distances.agr	PAPER_DEFINED	1.235	0.562	1.63	0.538	NORMALIZE LATER QUALITATIVE ONLY
QCD_large_distances.agr	PAPER_DEFINED	1.267	0.442	1.57	1.272	NORMALIZE LATER QUALITATIVE ONLY
QCD_large_distances.agr	PAPER_DEFINED	1.368	0.196	0.65	4.171	EXCLUDE FROM INTERPRETATION FOR NOW
beta6.3942dist789.agr	PAPER_DEFINED	1.368	0.253	0.65	4.171	EXCLUDE FROM INTERPRETATION FOR NOW

4.1 Review profile overlay



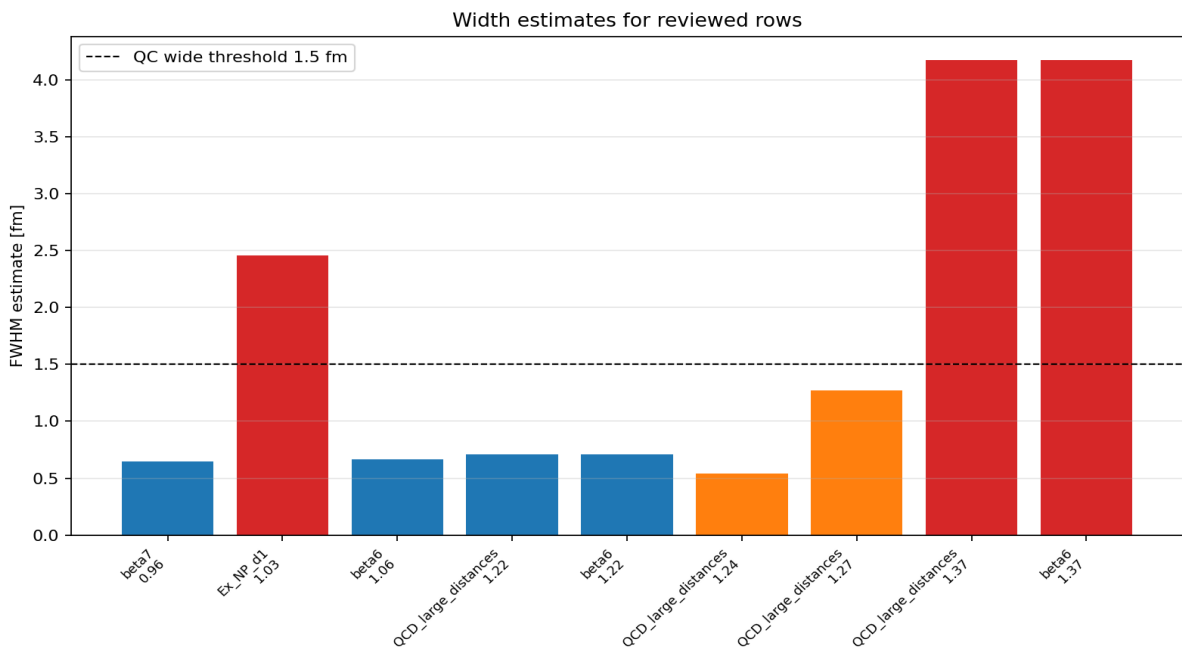
Reviewed rows shown separately. They are not mixed into the trusted trend.

4.2 Review signal-to-noise



Large-distance rows with low SNR remain qualitative-only or excluded.

4.3 Review width flags



Extreme widths or boundary maxima are not interpreted as physical route-width measurements.

5. UNNS mapping extracted

Physics quantity	UNNS interpretation
source separation d	route-extension coordinate
transverse $E_x(x_t; d)$ profile	localized route geometry
peak $E_x(d)$	central route intensity proxy
$\text{FWHM}(d)$	transverse route-width proxy
area under profile	routed field-strength proxy
trusted FULL / NP rows	primary localized-route samples

Physics quantity	UNNS interpretation
PAPER_DEFINED / large-distance rows	separate qualitative/control evidence class

localized color route persists across separation -> central intensity decreases within trusted interval -> transverse width increases moderately within trusted interval -> route geometry is redistributed, not freely externalized

6. Supported and not yet supported

Supported: the Baker profile data can be separated into a trusted FULL/NP trend set and a separate qualitative/control class; the trusted set provides a usable localized-route geometry over $d = 0.723\text{-}1.060\text{ fm}$; over that interval, peak field decreases and FWHM increases moderately with separation.

Not yet supported: a final fitted UNNS route-broadening law; a claim that flux-tube broadening reaches or explains string breaking; direct mixing of trusted FULL/NP profiles with PAPER_DEFINED or large-distance profiles; a derivation of QCD confinement from UNNS.

7. Correct wording for regime synthesis

The Baker flux-tube profiles provide a separation-dependent geometry for color-route localization. Within the trusted FULL/NP profile class, increasing source separation from approximately 0.72 to 1.06 fm is associated with reduced central field intensity and moderate transverse broadening. In UNNS terms, this supports treating the flux tube as a localized boundary-route whose geometry redistributes under route extension. The reviewed large-distance and paper-defined rows remain a separate qualitative evidence class until their normalization conventions are reconciled.

Avoid: UNNS derives QCD confinement or proves a universal flux-tube broadening law.

8. Placement

Place this Markdown and PDF in: color_confinement_data_seed_v0_1_1_provenance/reports/.

UNNS_Color_Confinement_Flux_Tube_Extension_Report.md
 UNNS_Color_Confinement_Flux_Tube_Extension_Report.pdf